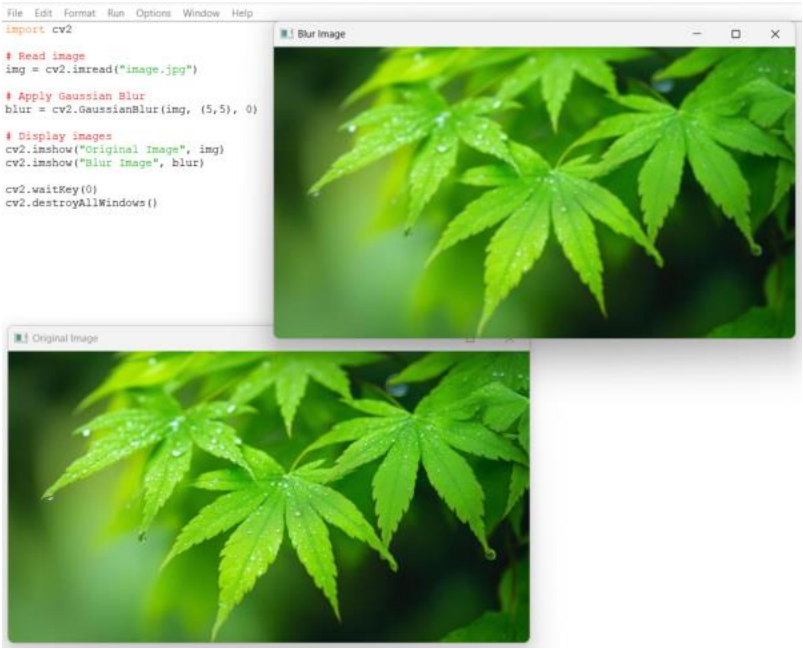
LAB EXPERIMENTS

NAME: K.DINESH REG NO: 192425125
COURSE CODE: ITA0503 COMPUTER VISION

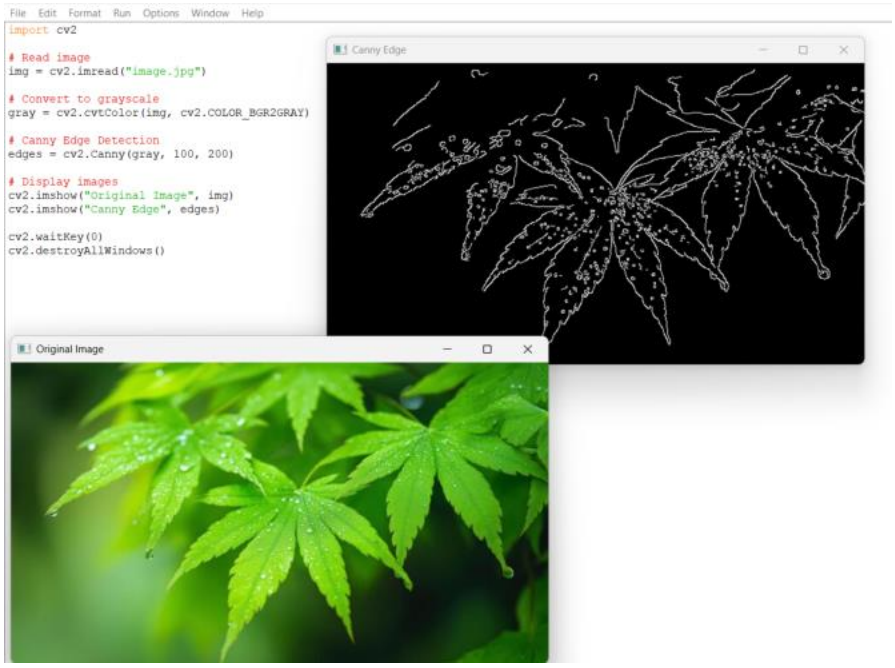
1.Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to Gray-scale.



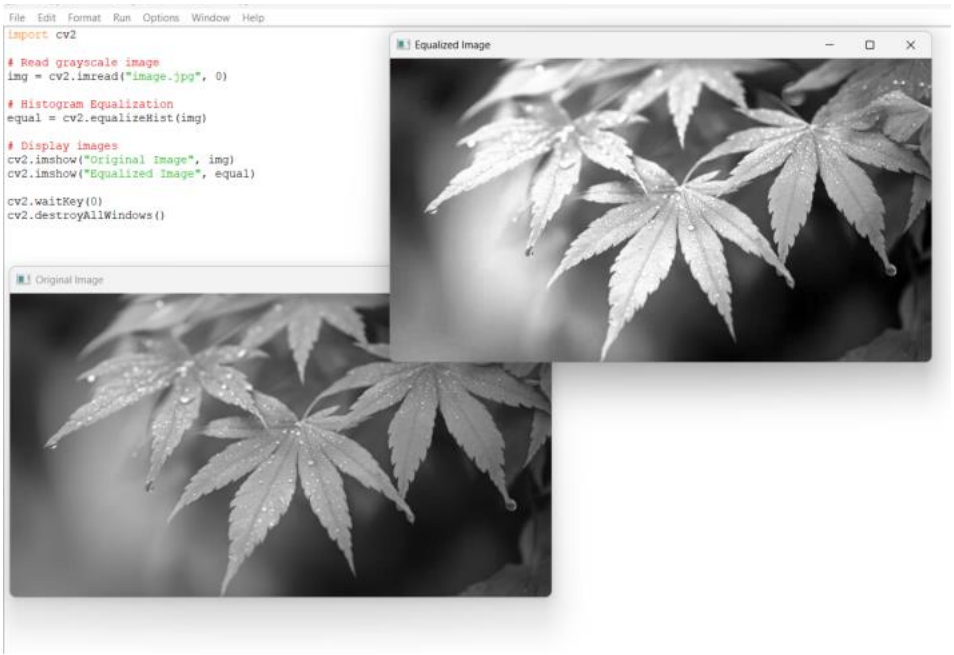
2. Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to Blur using Gaussian Blur



3.Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to show outline using Canny function.



4.Implement histogram equalization on the given image and compare it with the original image using Open CV.



5.Write a Python function to analyze the histogram of the given input image based on color levels using Open CV.

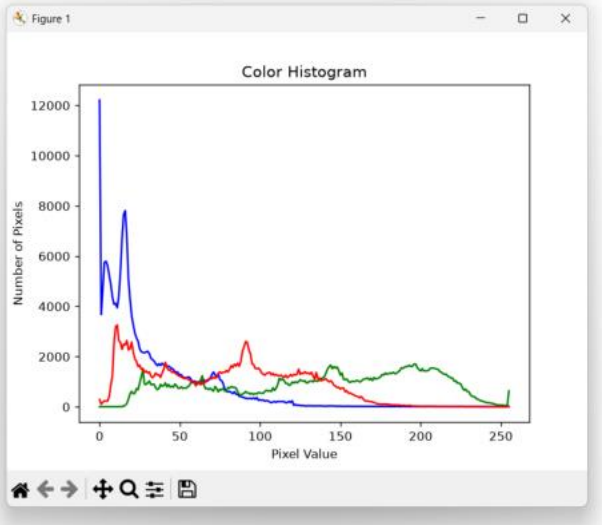
```
File  Edit  Format  Run  Options  Window  Help
import cv2
from matplotlib import pyplot as plt

# Read image
img = cv2.imread("image.jpg")

colors = ('b', 'g', 'r')

for i, col in enumerate(colors):
    hist = cv2.calcHist([img], [i], None, [256], [0,256])
    plt.plot(hist, color=col)

plt.title("Color Histogram")
plt.xlabel("Pixel Value")
plt.ylabel("Number of Pixels")
plt.show()
```



6.Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to erode using Erode function.

```
File  Edit  Format  Run  Options  Window  Help
import cv2
import numpy as np

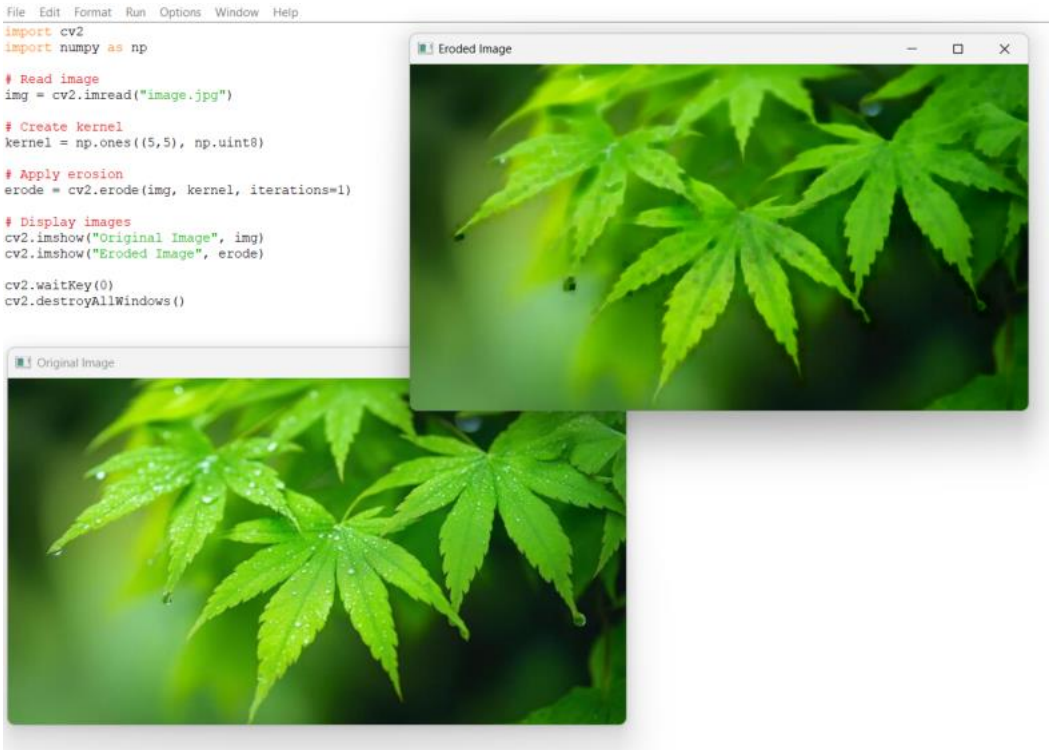
# Read image
img = cv2.imread("image.jpg")

# Create kernel
kernel = np.ones((5,5), np.uint8)

# Apply erosion
erode = cv2.erode(img, kernel, iterations=1)

# Display images
cv2.imshow("Original Image", img)
cv2.imshow("Eroded Image", erode)

cv2.waitKey(0)
cv2.destroyAllWindows()
```



7.Perform basic video processing operations on the captured video. Read captured video in python and display the video, in slow motion and in fast motion.

```
import cv2

# Read video
cap = cv2.VideoCapture(r"C:\Users\Kaviya V\Downloads\video.webm")

if not cap.isOpened():
    print("Cannot open video")
    exit()

print("Choose Playback Speed")
print("1 - Slow Motion")
print("2 - Normal")
print("3 - Fast Motion")

choice = input("Enter your choice: ")

if choice == "1":
    delay = 100      # Slow Motion
elif choice == "3":
    delay = 10       # Fast Motion
else:
    delay = 30       # Normal Speed

while True:
    ret, frame = cap.read()

    if not ret:
        break

    cv2.imshow("Video Playback", frame)

    if cv2.waitKey(delay) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```

```
*IDLE Shell 3.14.3*
File  Edit  Shell  Debug  Options  Window  Help
Enter "help" below or click "Help" abo
ve for more information.
>>>
=== RESTART: C:\Users\Kaviya V\OneDriv
e\Documents\A~COLLEGE\CV\EXPS\EXP 7.py
===
Choose Playback Speed
1 - Slow Motion
2 - Normal
3 - Fast Motion
Enter your choice: 1
Ln: 10  Col: 0
```



8.Perform basic Image Handling and processing operations on the Image is to read an Image in python and Dilate an Image using Dilate function

```
File  Edit  Format  Run  Options  Window  Help
import cv2
import numpy as np

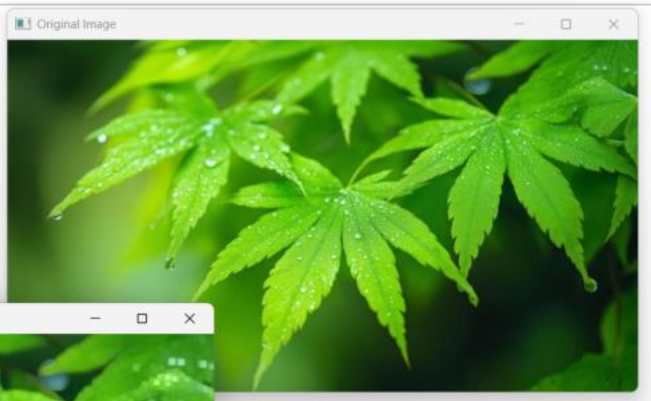
# Read image
img = cv2.imread("image.jpg")

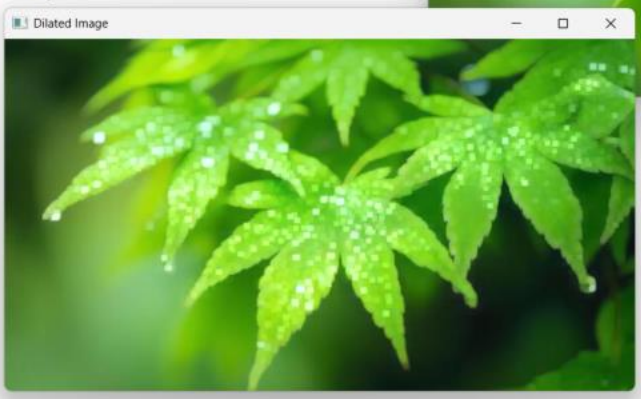
# Create kernel
kernel = np.ones((5,5), np.uint8)

# Apply dilation
dilate = cv2.dilate(img, kernel, iterations=1)

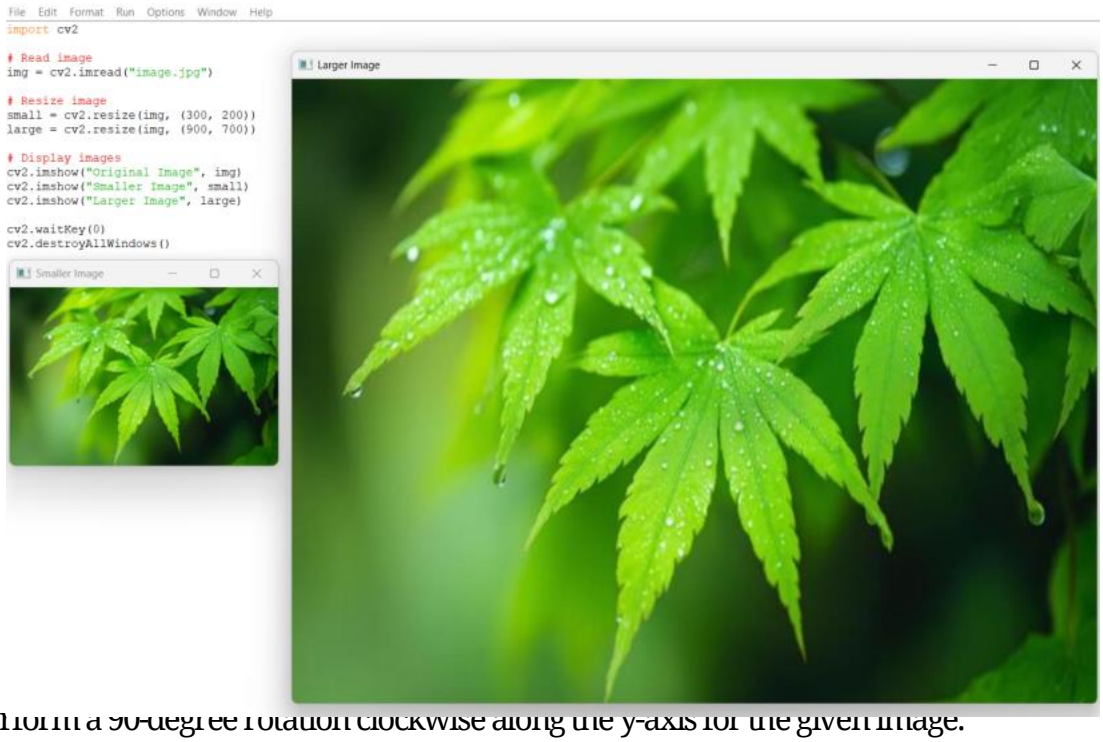
# Display images
cv2.imshow("Original Image", img)
cv2.imshow("Dilated Image", dilate)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

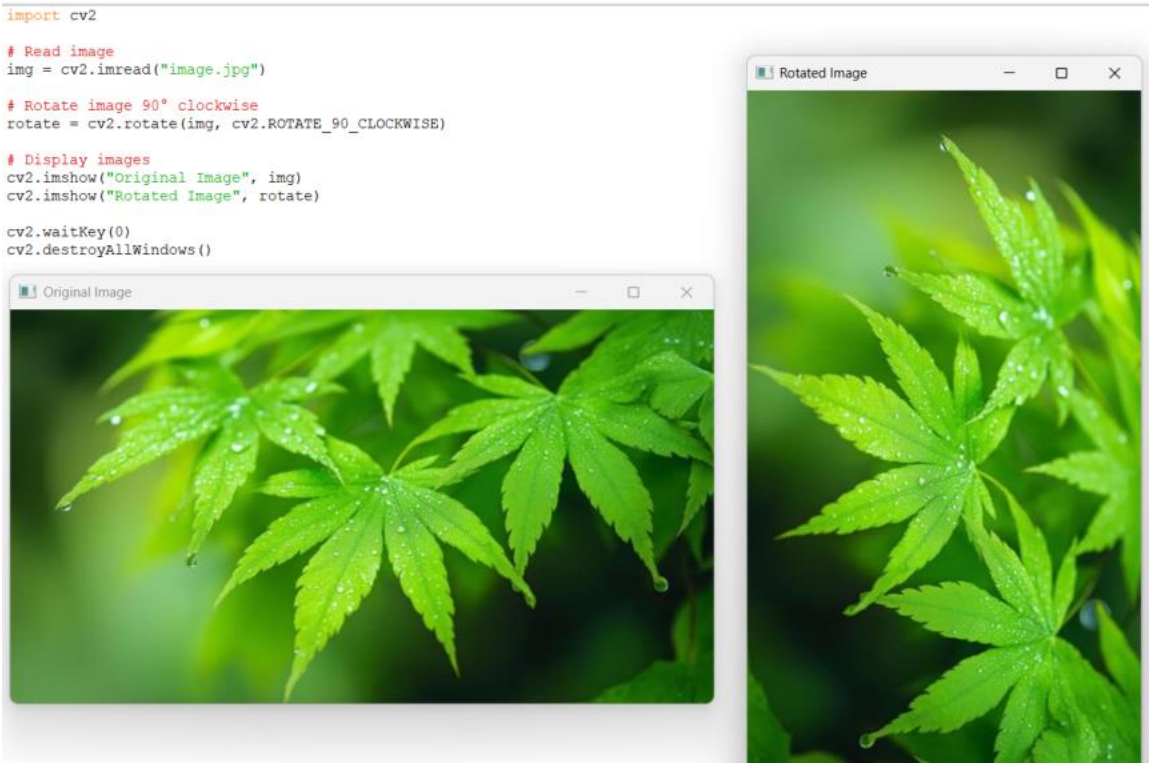




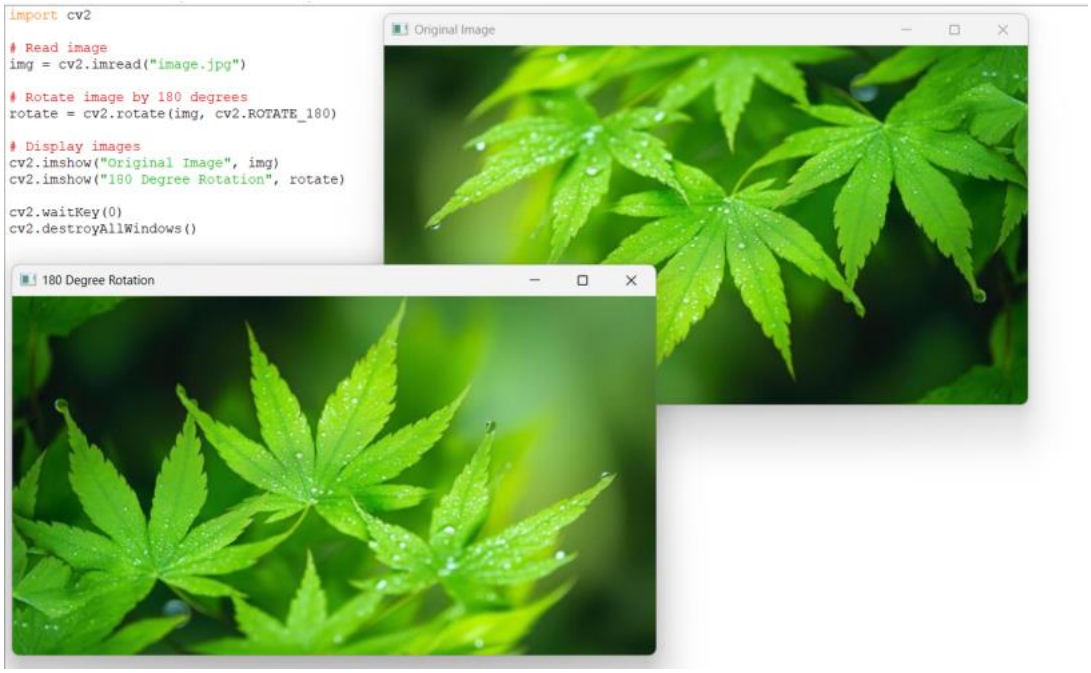
9.Implement the image scaling techniques to resize the images to both bigger and small sizes



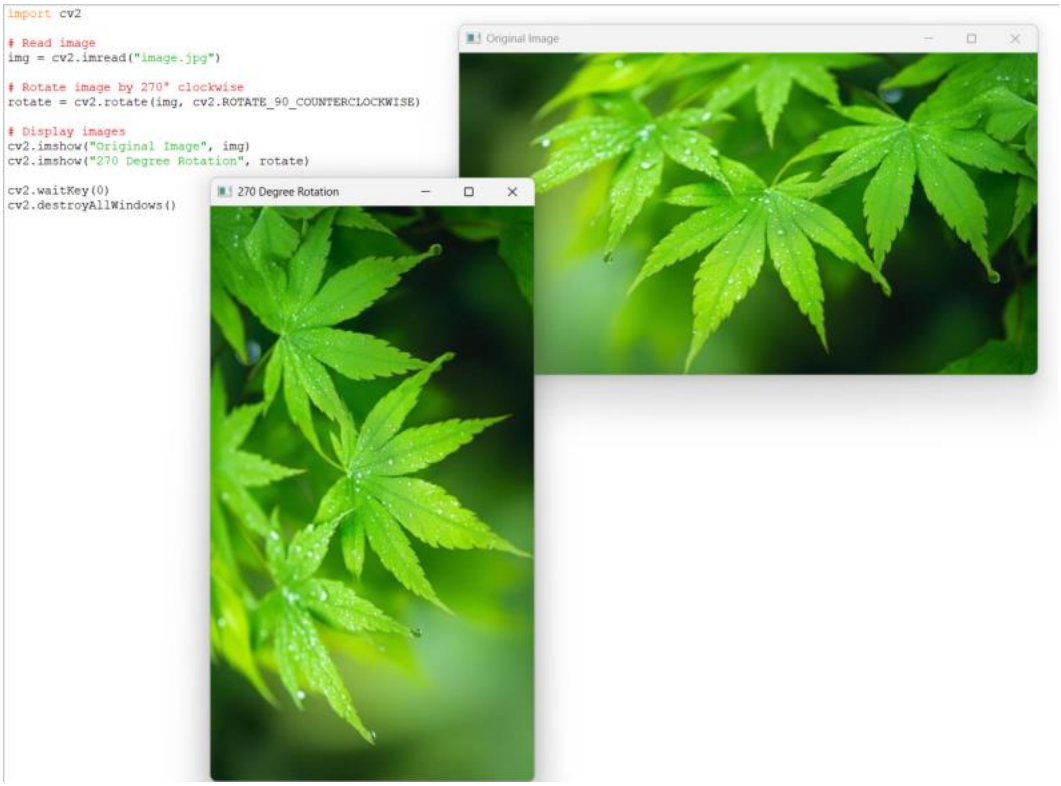
10.Perform a 90-degree rotation clockwise along the y-axis for the given image.



11 .Perform a 180-degree rotation clockwise along the y-axis for the given image.



12.Perform a 270-degree rotation clockwise along the y-axis for the given image.



13.Perform Affine Transformation on the given image using python and Open CV

```
import cv2
import numpy as np

# Read image
img = cv2.imread("image.jpg")

rows, cols = img.shape[:2]

# Source points
pts1 = np.float32([[50,50],[200,50],[50,200]])

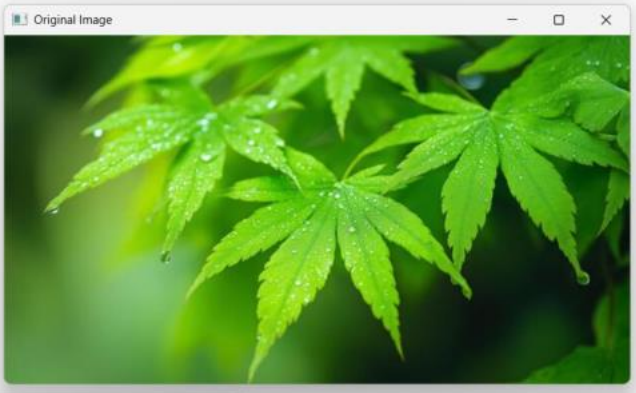
# Destination points
pts2 = np.float32([[10,100],[200,50],[100,250]])

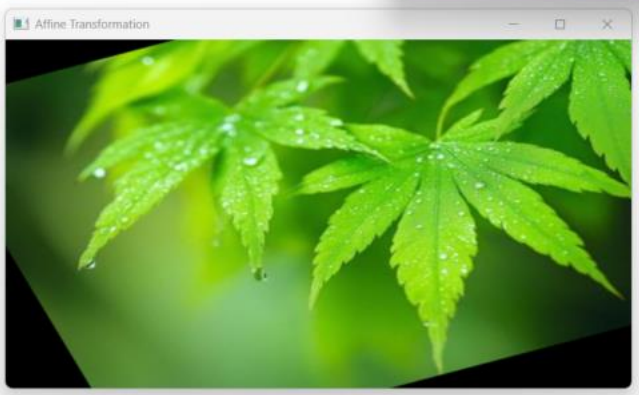
# Affine transformation
matrix = cv2.getAffineTransform(pts1, pts2)

result = cv2.warpAffine(img, matrix, (cols, rows))

cv2.imshow("Original Image", img)
cv2.imshow("Affine Transformation", result)

cv2.waitKey(0)
cv2.destroyAllWindows()
```





14.Perform Perspective Transformation on the given image using python and Open CV.

```
File  Edit  Format  Run  Options  Window  Help
import cv2
import numpy as np

# Read image
img = cv2.imread("image.jpg")

rows, cols = img.shape[:2]

# Source points
pts1 = np.float32([[50,50],[300,50],[50,300],[300,300]])

# Destination points
pts2 = np.float32([[0,0],[300,0],[100,300],[300,300]])

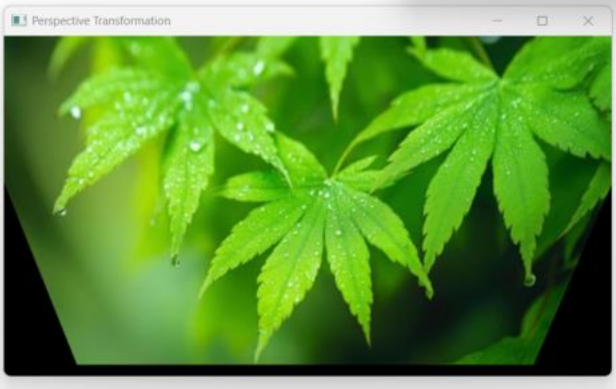
# Perspective transformation
matrix = cv2.getPerspectiveTransform(pts1, pts2)

result = cv2.warpPerspective(img, matrix, (cols, rows))

cv2.imshow("Original Image", img)
cv2.imshow("Perspective Transformation", result)

cv2.waitKey(0)
cv2.destroyAllWindows()
```





15.Perform basic Image Handling and processing operations on the image is to read an image in python and detect the corners in the image using Harris Corner Detection function

File Edit Format Run Options Window Help

```
import cv2
import numpy as np

# Read image
img = cv2.imread("image.jpg")

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
gray = np.float32(gray)

# Harris Corner Detection
corners = cv2.cornerHarris(gray, 2, 3, 0.04)

# Mark corners
img[corners > 0.01 * corners.max()] = [0, 0, 255]

cv2.imshow("Harris Corner Detection", img)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

